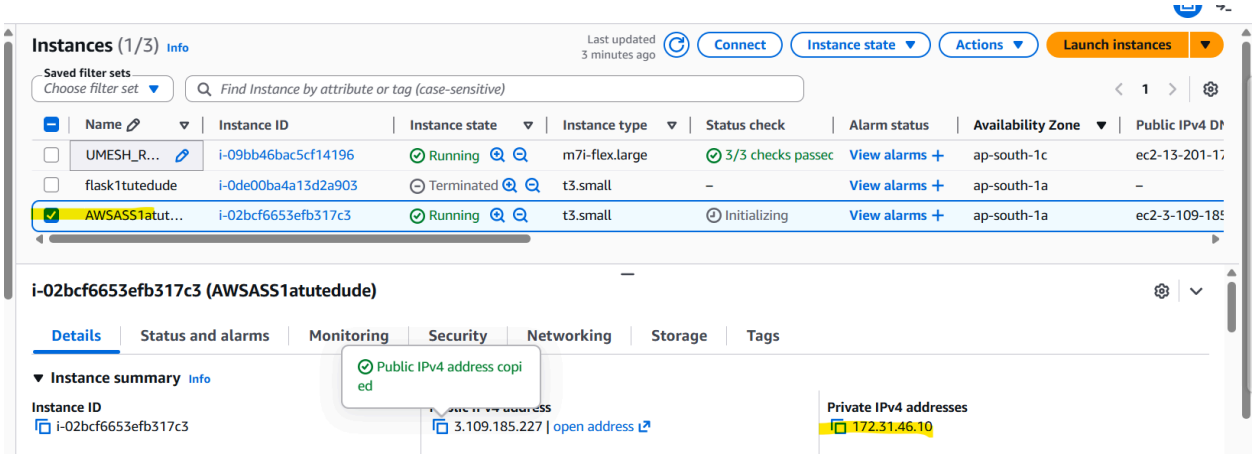
Note-: If you are new to AWS, you can create an AWS account and take advantage of the **one-year free tier**. This free tier provides access to a limited set of resources and services at no cost. For more details, you can refer to the official AWS Free Tier page at https://aws.amazon.com/free.

1. Deploy Your flask backend and express frontend in amazon single ec2 instance.

I have created instance using the AWS application



Connected to AWS server using CMD

CMD : **ssh -i assignment.pem ubuntu@3.109.185.227**

```
Applications   Terminal - ubuntu@ip-1...
                                                    Terminal - ubuntu@ip-172-31
File  Edit  View  Terminal  Tabs  Help
ubuntu@ip-172-31-16-42:~/Downloads$ ssh -i assignment.pem ubuntu@3.109.185.227
The authenticity of host '3.109.185.227 (3.109.185.227)' can't be established.
ED25519 key fingerprint is SHA256:a0UFELkAM08TpRKRIceJYWsAmQDgN4Bz6kjZBGnfAdA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.109.185.227' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Apr 22 04:55:35 UTC 2026

  System load:  0.19              Temperature:           -273.1 C
  Usage of /:   28.4% of 6.71GB   Processes:             149
  Memory usage: 14%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.46.10


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Wed Apr 22 04:53:35 2026 from 13.233.177.4
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-46-10:~$ ^C
ubuntu@ip-172-31-46-10:~$
```

Updating packages our server :

CMD : **sudo apt update**

CMD : **sudo apt upgrade -y**

Install Python, Node.js, Git, and Nginx :

CMD : **sudo apt install python3-pip python3-venv git -y**

CMD : **sudo apt install nodejs npm -y**

CMD : **sudo apt install nginx -y**

```
ubuntu@ip-172-31-46-10:~$ python3 --version
Python 3.12.3
ubuntu@ip-172-31-46-10:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-46-10:~$ node -v
v18.19.1
ubuntu@ip-172-31-46-10:~$ niginx -V
Command 'niginx' not found, did you mean:
  command 'nginx' from deb nginx (1.24.0-2ubuntu7.6)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-46-10:~$ nginx -V
nginx version: nginx/1.24.0 (Ubuntu)
built with OpenSSL 3.0.13 30 Jan 2024
TLS SNI support enabled
configure arguments: --with-cc-opt='-g -O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/nginx-gpkOWo/nginx-1.24.0=. -flto=auto -ffat-lt
ector-strong -fstack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/build/nginx-gpkOWo/nginx-1.24.0=/usr/src/nginx-1.24.0-2ubuntu
-D_FORTIFY_SOURCE=3' --with-ld-opt='-Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -Wl,-z,relro -Wl,-z,now -fPIC' --prefix=/usr/share/nginx --conf-path=/etc/ngin
g-path=/var/log/nginx/access.log --error-log-path=stderr --lock-path=/var/lock/nginx.lock --pid-path=/run/nginx.pid --modules-path=/usr/lib/nginx/modules --http-client-
b/nginx/body --http-fastcgi-temp-path=/var/lib/nginx/fastcgi --http-proxy-temp-path=/var/lib/nginx/proxy --http-scgi-temp-path=/var/lib/nginx/scgi --http-uwsgi-temp-pat
 --with-compat --with-debug --with-pcre-jit --with-http_ssl_module --with-http_stub_status_module --with-http_realip_module --with-http_auth_request_module --with-http_
dav_module --with-http_slice_module --with-threads --with-http_addition_module --with-http_flv_module --with-http_gunzip_module --with-http_gzip_static_module --with-ht
tp_random_index_module --with-http_secure_link_module --with-http_sub_module --with-mail_ssl_module --with-stream_ssl_module --with-stream_ssl_preread_module --with-st
ith-http_geoip_module=dynamic --with-http_image_filter_module=dynamic --with-http_perl_module=dynamic --with-http_xslt_module=dynamic --with-mail=dynamic --with-stream=
geoip_module=dynamic
ubuntu@ip-172-31-46-10:~$
```

Getting files from the GIT using below commands:

CMD : **git clone https://github.com/UmeshChandra2001/DockerP1.git**

```
ubuntu@ip-172-31-46-10:~/awsassignment1$ git clone https://github.com/UmeshChandra2001/DockerP1.git
Cloning into 'DockerP1'...
remote: Enumerating objects: 23, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 23 (delta 2), reused 13 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (23/23), 1.16 MiB | 3.16 MiB/s, done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-46-10:~/awsassignment1$ ls -al
total 16
drwxrwxr-x 4 ubuntu ubuntu 4096 Apr 22 06:16 .
drwxr-x--- 5 ubuntu ubuntu 4096 Apr 22 06:12 ..
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 22 06:15 Backend
drwxrwxr-x 5 ubuntu ubuntu 4096 Apr 22 06:17 DockerP1
ubuntu@ip-172-31-46-10:~/awsassignment1$ cd DockerP1/
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1$ ls -al
total 1384
drwxrwxr-x 5 ubuntu ubuntu    4096 Apr 22 06:17  .
drwxrwxr-x 4 ubuntu ubuntu    4096 Apr 22 06:16  ..
drwxrwxr-x 8 ubuntu ubuntu    4096 Apr 22 06:17  .git
-rw-rw-r-- 1 ubuntu ubuntu      42 Apr 22 06:17  .gitignore
drwxrwxr-x 2 ubuntu ubuntu    4096 Apr 22 06:17  Backend
-rw-rw-r-- 1 ubuntu ubuntu    3216 Apr 22 06:17 'Excuted cmds.bat'
drwxrwxr-x 3 ubuntu ubuntu    4096 Apr 22 06:17  Frontend
-rw-rw-r-- 1 ubuntu ubuntu 1377633 Apr 22 06:17 'P V UMESH CHANDRA DOCKER ASSIGNMENT.pdf'
-rw-rw-r-- 1 ubuntu ubuntu      38 Apr 22 06:17  README.md
-rwxrwxr-x 1 ubuntu ubuntu     266 Apr 22 06:17  docker-compose.yaml
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1$ cd Backend/
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Backend$ ls -al
total 20
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 22 06:17 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Apr 22 06:17 ..
-rwxrwxr-x 1 ubuntu ubuntu  150 Apr 22 06:17 Dockerfile
-rwxrwxr-x 1 ubuntu ubuntu  303 Apr 22 06:17 app.py
-rwxrwxr-x 1 ubuntu ubuntu    5 Apr 22 06:17 requirements.txt
```

Create virtual environment & install requirements

CMD : **python3 -m venv venv**

CMD : **source venv/bin/activate**

CMD : **pip install -r requirements.txt**

```
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1$ cd Backend/
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Backend$ python3 -m venv venv
ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Backend$ source venv/bin/activate
(venv) ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Backend$ pip install -r requirements.txt
Collecting flask (from -r requirements.txt (line 1))
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask->-r requirements.txt (line 1))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask->-r requirements.txt (line 1))
  Downloading click-8.3.2-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting markupsafe>=2.1.1 (from flask->-r requirements.txt (line 1))
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting werkzeug>=3.1.0 (from flask->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 kB 6.7 MB/s eta 0:00:00
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.3.2-py3-none-any.whl (108 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.4/108.4 kB 9.6 MB/s eta 0:00:00
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 13.1 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 20.0 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, j
Successfully installed blinker-1.9.0 click-8.3.2 flask-3.1.3 itsdangerous-2.2.0 jinj
-3.0.3 werkzeug-3.1.8
(venv) ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Backend$ cd ..
```

File Manager
Browse the file system

Installing npm for frontend

CMD : **npm install**

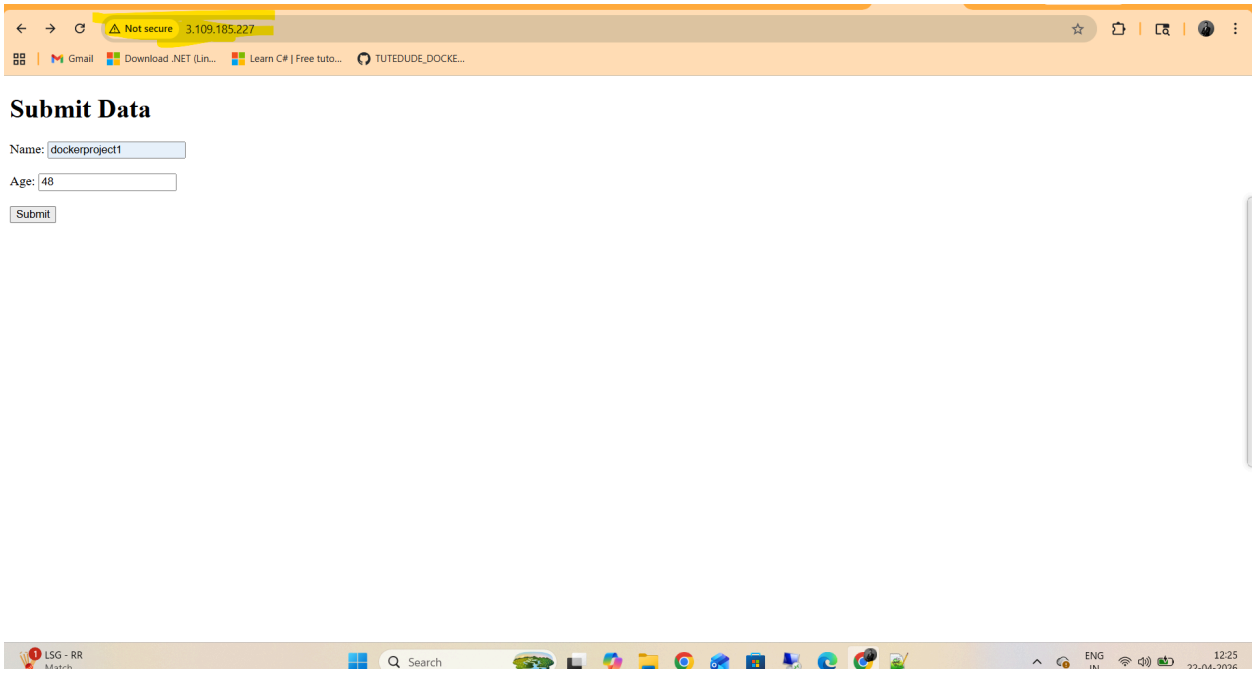
```
(venv) ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Frontend$ npm install

added 77 packages, and audited 78 packages in 3s

17 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
(venv) ubuntu@ip-172-31-46-10:~/awsassignment1/DockerP1/Frontend$
```

CONNECT NGINX

server {

    listen 80;

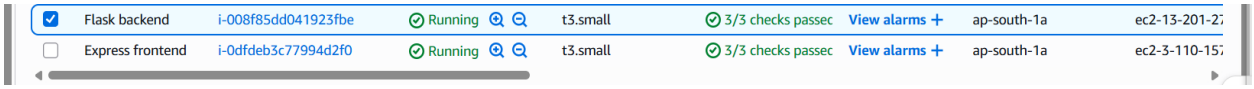
```
    # Flask backend
    location /api {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }

    # Express frontend
    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
```


Not secure
3.109.185.227
Gmail
Download .NET (Lin...
Learn C# | Free tuto...
TUTEDUDE_DOCKE...
Submit Data
Name: dockerproject1
Age: 48
Submit
LSG - RR
Search
ENG
12:25

2. Deploy Your flask backend and express frontend in separate ec2 instances.

ANS :

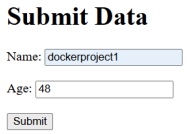
Create one EC2 instance for **Flask backend** and another for **Express frontend**.



Flask backend	i-008f85dd041923fbe	Running	t3.small	3/3 checks passed	View alarms +	ap-south-1a	ec2-13-201-27
Express frontend	i-0dfdeb3c77994d2f0	Running	t3.small	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-110-157

Deployed frontend code in one server and backend code in one severe

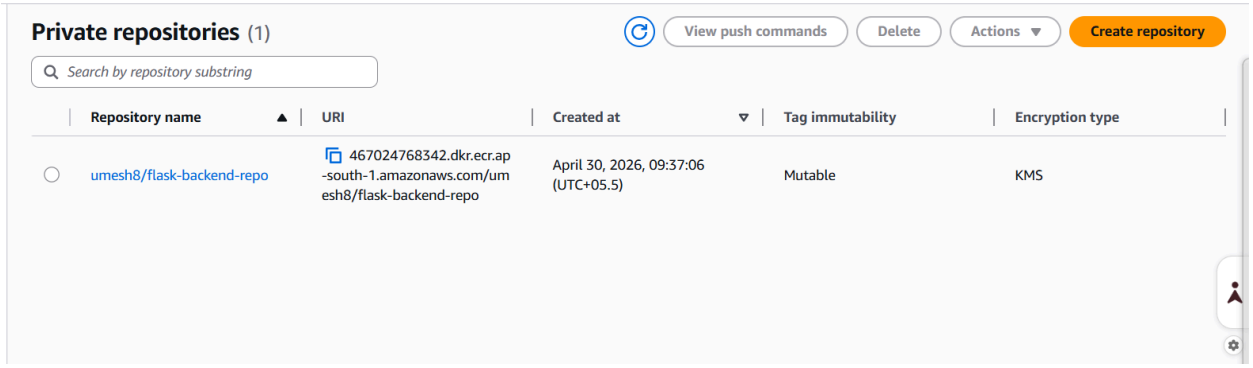
Final output



3. Deploy Your flask backend and express frontend Docker Container using aws ecr, ecs and vpc services

ANS:

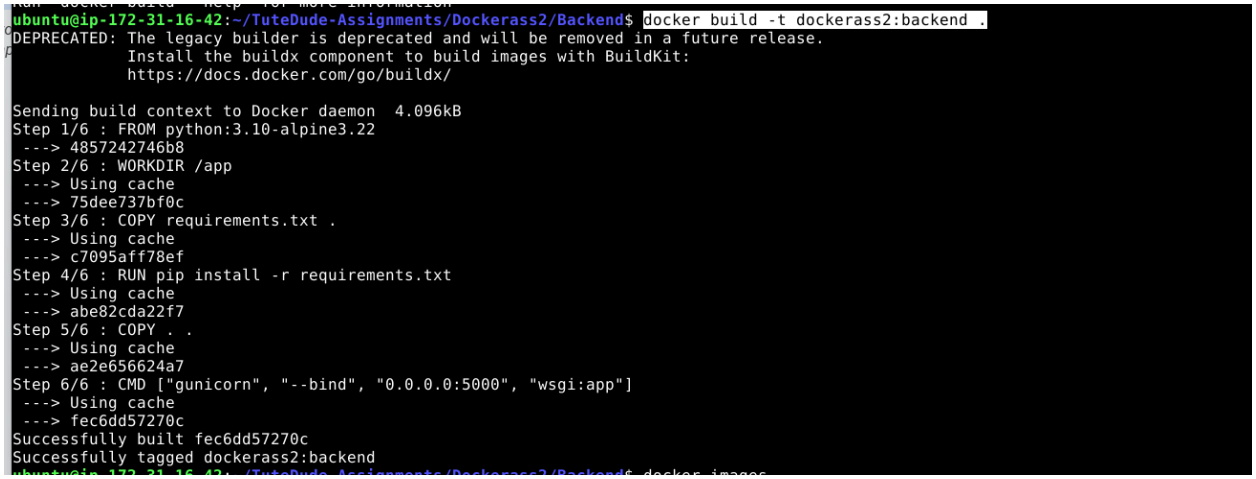
Created one ECR called flask-backend-repo for the backend



Connect the repository using AWS IAM access IDs.

Pushing code docker

CMD : **docker build -t dockerass2:backend .**



```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker build -t dockerass2:backend .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/6 : FROM python:3.10-alpine3.22
 ---> 4857242746b8
Step 2/6 : WORKDIR /app
 ---> Using cache
 ---> 75dee737bf0c
Step 3/6 : COPY requirements.txt .
 ---> Using cache
 ---> c7095aff78ef
Step 4/6 : RUN pip install -r requirements.txt
 ---> Using cache
 ---> abe82cda22f7
Step 5/6 : COPY . .
 ---> Using cache
 ---> ae2e656624a7
Step 6/6 : CMD ["gunicorn", "--bind", "0.0.0.0:5000", "wsgi:app"]
 ---> Using cache
 ---> fec6dd57270c
Successfully built fec6dd57270c
Successfully tagged dockerass2:backend
```

After building, check

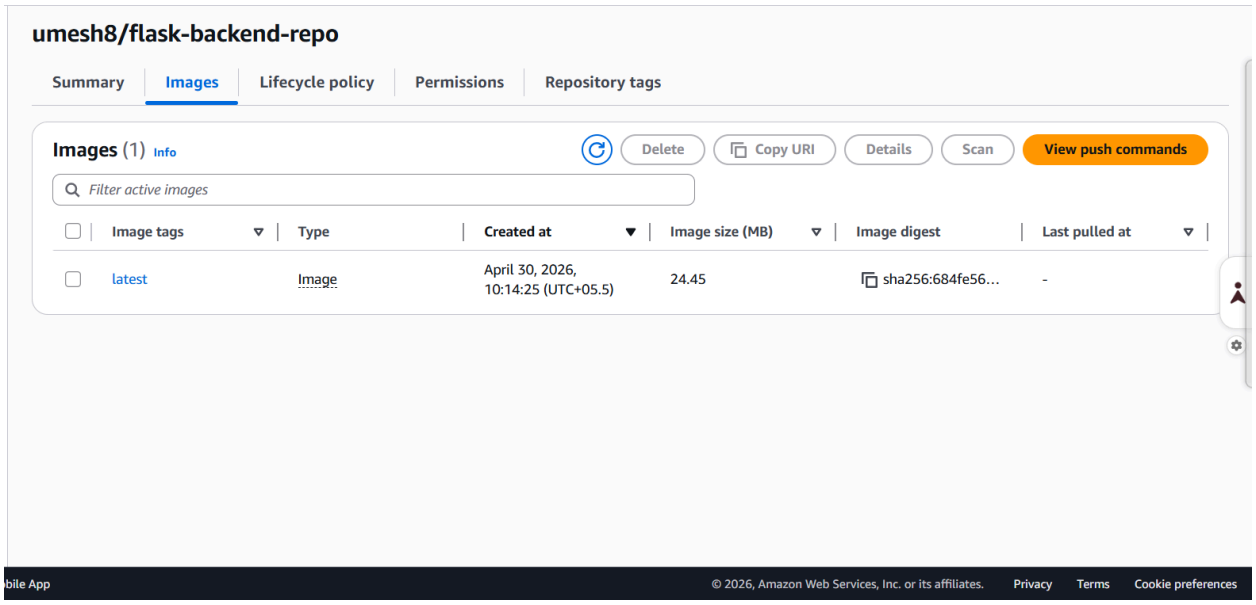
CMD : **docker images**

Once the image is built pushing ecr repository:

CMD : **docker tag dockerass2:backend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo:latest**

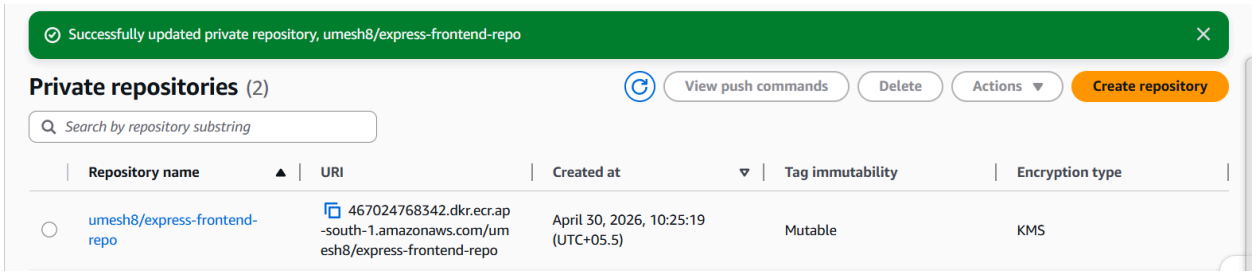
CMD : **docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo:latest**

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker images
REPOSITORY                                  TAG                 IMAGE ID       CREATED        SIZE
dockerass2_frontend                         latest              c68412a838d7   22 hours ago   148MB
dockerass2                                  backend             fec6dd57270c   22 hours ago   63.3MB
dockerass2_backend                          latest              fec6dd57270c   22 hours ago   63.3MB
dockerp1_frontend                           latest              ea0b48fecc45   3 weeks ago    148MB
umeshchandra2001/dockerproject1/frontend    latest              ea0b48fecc45   3 weeks ago    148MB
umeshchandra2001/tdproject1                 fronten             ea0b48fecc45   3 weeks ago    148MB
dockerp1_backend                            latest              e399f8a0e484   3 weeks ago    63.3MB
umeshchandra2001/tdproject1                 backend             e399f8a0e484   3 weeks ago    63.3MB
node                                        20.20.2-alpine3.23  ccfb9f0dc7ac   4 weeks ago    136MB
python                                      3.10-alpine3.22     4857242746b8   8 weeks ago    51.4MB
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker tag dockerass2:backend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/flask-backend-repo:latest
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/flask-backend-repo:latest
The push refers to repository [467024768342.dkr.ecr.ap-south-1.amazonaws.com/flask-backend-repo]
2ff23f8aadc0: Preparing
ce400c3355a5: Preparing
c337b36c31dd: Preparing
013bce42f61e: Preparing
cf43eeae7c5d: Preparing
7ea0976f4b2e: Waiting
fe81b6fa373b: Waiting
cce92674e987: Waiting
name unknown: The repository with name 'flask-backend-repo' does not exist in the registry with id '467024768342'
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker tag dockerass2:backend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo:
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$ docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo:latest
The push refers to repository [467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo]
2ff23f8aadc0: Pushed
ce400c3355a5: Pushed
c337b36c31dd: Pushed
013bce42f61e: Pushed
cf43eeae7c5d: Pushed
7ea0976f4b2e: Pushed
fe81b6fa373b: Pushed
cce92674e987: Pushed
latest: digest: sha256:684fe56494b6d2bc5467044d4a5a01267923515a5386976395bc0285843d3296 size: 1988
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Backend$
```



SAME DOING FOR FRONTEND ALSO:

Created one more ECR called express-frontend-repo for the frontend



CMD :

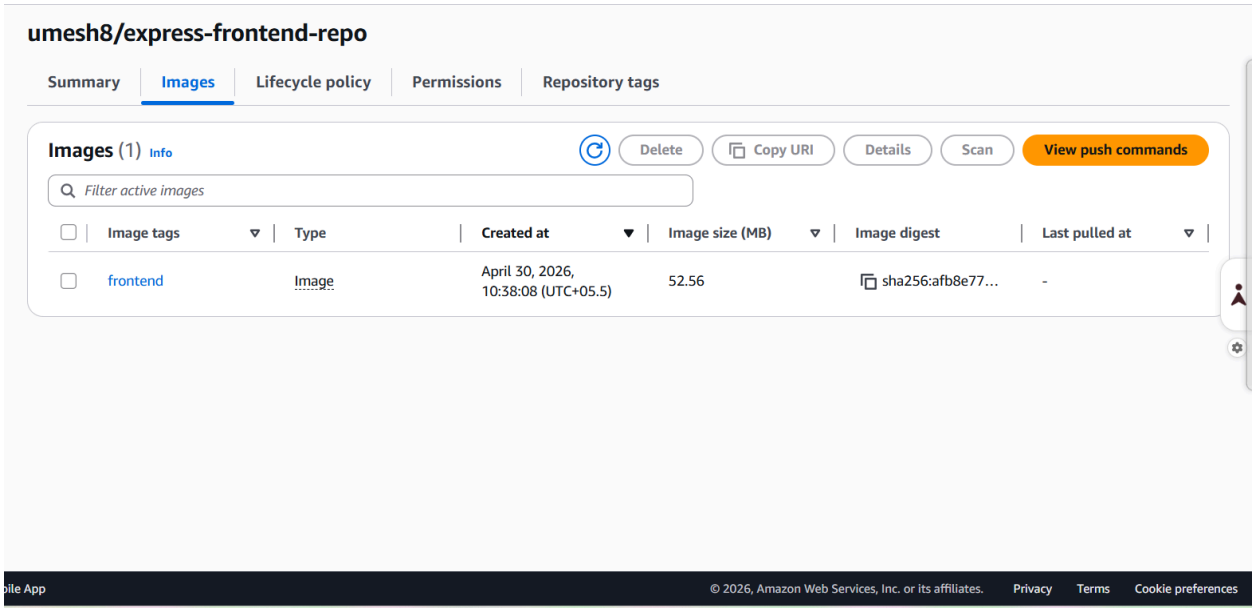
docker build -t dockerass2:frontend .

docker images

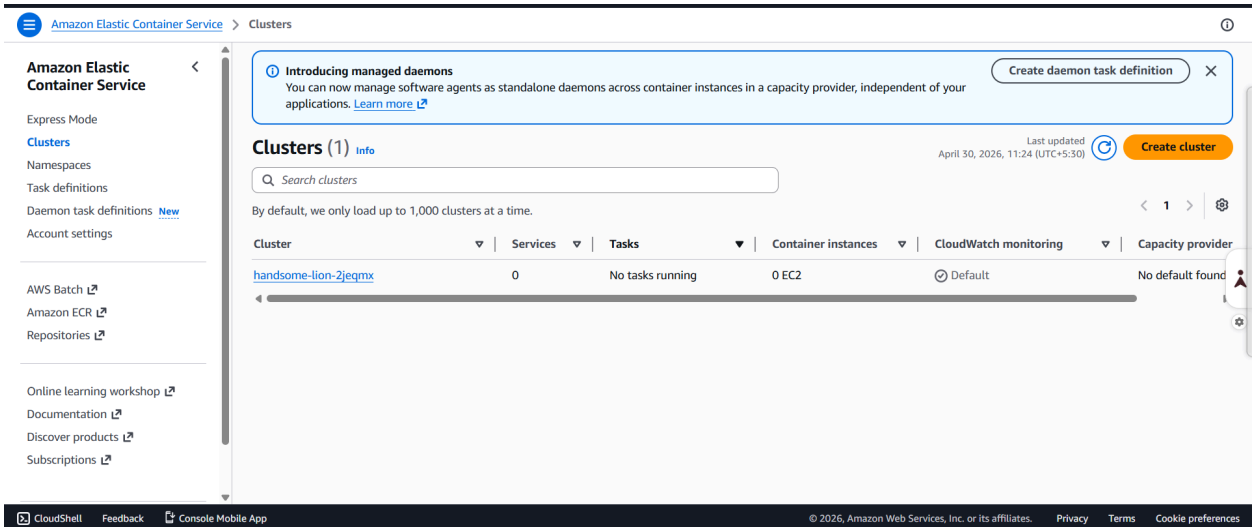
docker tag dockerass2:frontend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo :frontend

docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo :frontend

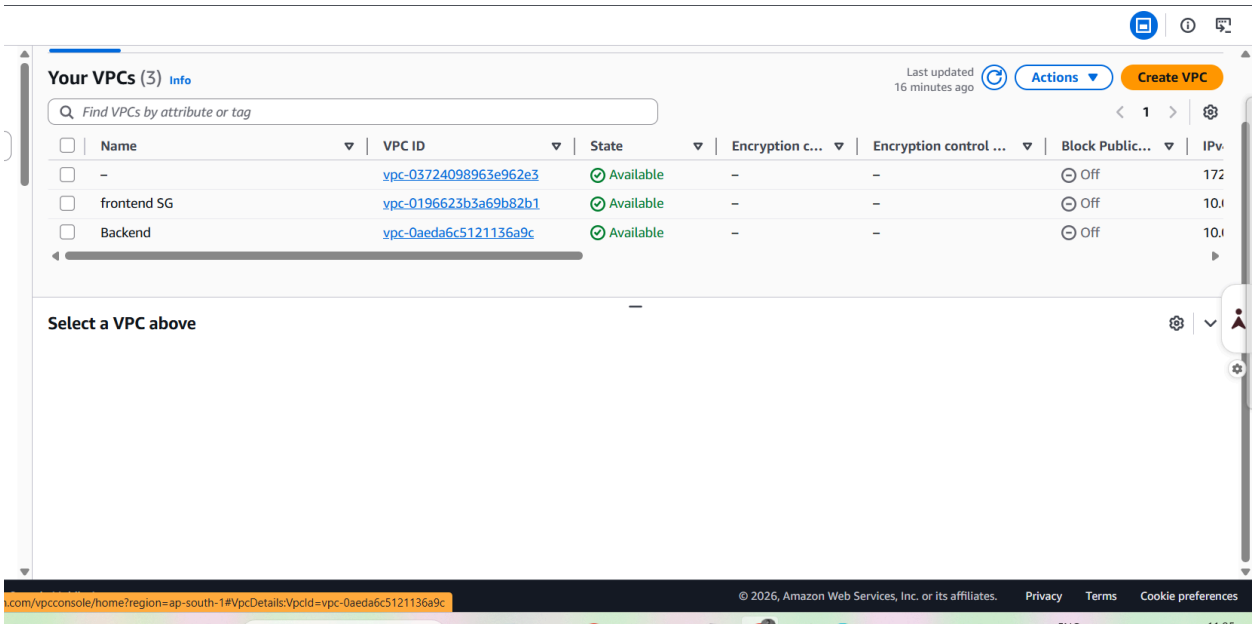
```
 ---> Using cache
 ---> 04515fd920d2
Step 6/7 : EXPOSE 3000
 ---> Using cache
 ---> 05bab3db4dd0
Step 7/7 : CMD ["node", "server.js"]
 ---> Using cache
 ---> c68412a838d7
Successfully built c68412a838d7
Successfully tagged dockerass2:frontend
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$ docker images
REPOSITORY                                                          TAG                 IMAGE ID       CREATED        SIZE
dockerass2                                                          frontend            c68412a838d7   23 hours ago   148MB
dockerass2_frontend                                                 latest              c68412a838d7   23 hours ago   148MB
dockerass2_backend                                                  latest              fec6dd57270c   23 hours ago   63.3MB
umeshchandra2001/dockerass2                                         backend             fec6dd57270c   23 hours ago   63.3MB
467024768342.dkr.ecr.ap-south-1.amazonaws.com/flask-backend-repo    latest              fec6dd57270c   23 hours ago   63.3MB
467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/flask-backend-repo   latest       fec6dd57270c   23 hours ago   63.3MB
dockerass2                                                          backend             fec6dd57270c   23 hours ago   63.3MB
dockerp1_frontend                                                   latest              ea0b48fecc45   3 weeks ago    148MB
umeshchandra2001/dockerproject1/frontend                            latest              ea0b48fecc45   3 weeks ago    148MB
umeshchandra2001/tdproject1                                         fronten             ea0b48fecc45   3 weeks ago    148MB
dockerp1_backend                                                    latest              e399f8a0e484   3 weeks ago    63.3MB
umeshchandra2001/tdproject1                                         backend             e399f8a0e484   3 weeks ago    63.3MB
node                                                                20.20.2-alpine3.23  ccfb9f0dc7ac   4 weeks ago    136MB
python                                                              3.10-alpine3.22     4857242746b8   8 weeks ago    51.4MB
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$ docker tag dockerass2:frontend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$ docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo:latestfrontend
The push refers to repository [467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo]
tag does not exist: 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo:latestfrontend
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$ docker tag dockerass2:frontend 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$ docker push 467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo:frontend
The push refers to repository [467024768342.dkr.ecr.ap-south-1.amazonaws.com/umesh8/express-frontend-repo]
c2781183dd30: Pushed
9517ec7125ff: Pushed
5c2db8033810: Pushed
4c53c1aa63ca: Pushed
52caa5bd1d9d: Pushed
5906df279b58: Pushed
466f2827c5d1: Pushed
989e799e6349: Pushed
frontend: digest: sha256:afb8e779b4ee8c102121ae9793cffe3953e50a2bc9164ebe0fb1c04746b78ff7 size: 1990
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/Dockerass2/Frontend$
```



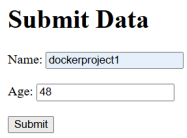
Created clustered based on ECR



CREATED VPC and Make application running



APPLICATION:



Submission guidelines -: Share your github repo link and your deployed app url(first message in the chat before making it live until then stop your instance so that you won't be cost much)